Robust Bayesian Inference under Partially Specified Priors

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Main idea

Economic reasoning may only **partially identify prior beliefs**, just as data may only partially identify structural parameters. I characterize **robust posterior bounds** over prior classes defined by defensible primitives and show that in **set-identified models** they can be computed using **linear programming**.

Intuition

Consider a simple consumption function

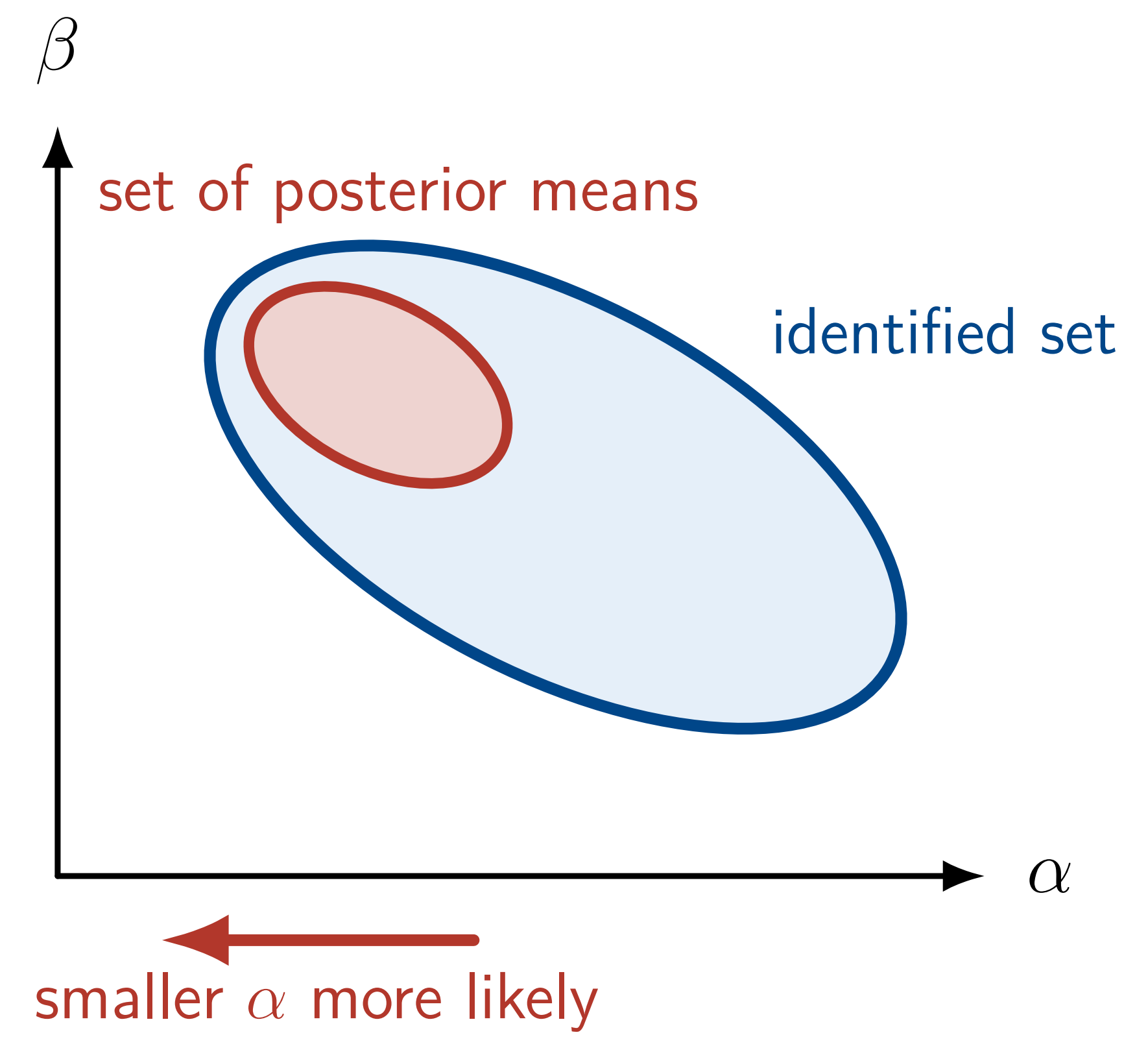
$$C_i = \alpha + \beta Y_i + \varepsilon_i,$$

where α is autonomous consumption and β the marginal propensity to consume (MPC).

The data and deterministic assumptions may identify only a **set of admissible parameters** (α, β). A frequentist reports this identified set.

A Bayesian may know more, but not enough to specify a unique prior - for example: **smaller values of autonomous consumption are more likely**.

This **shrinks the set of posterior conclusions** about the MPC:



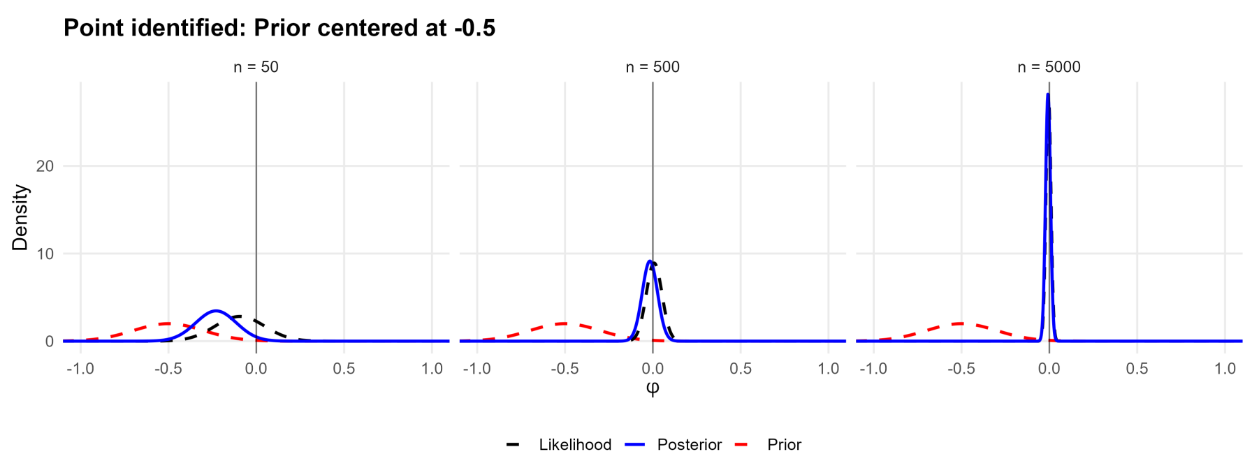
> **Weak prior information about one model feature, might inform other features.**

Why does this matter?

Consider the example of Moon and Schorfheide (2012). Let ϕ be $\mathbb{E}[Y]$ and we observe $Y_1, \dots, Y_n$ iid. Suppose theory only tells us $\phi \le \theta \le \phi + 2$.

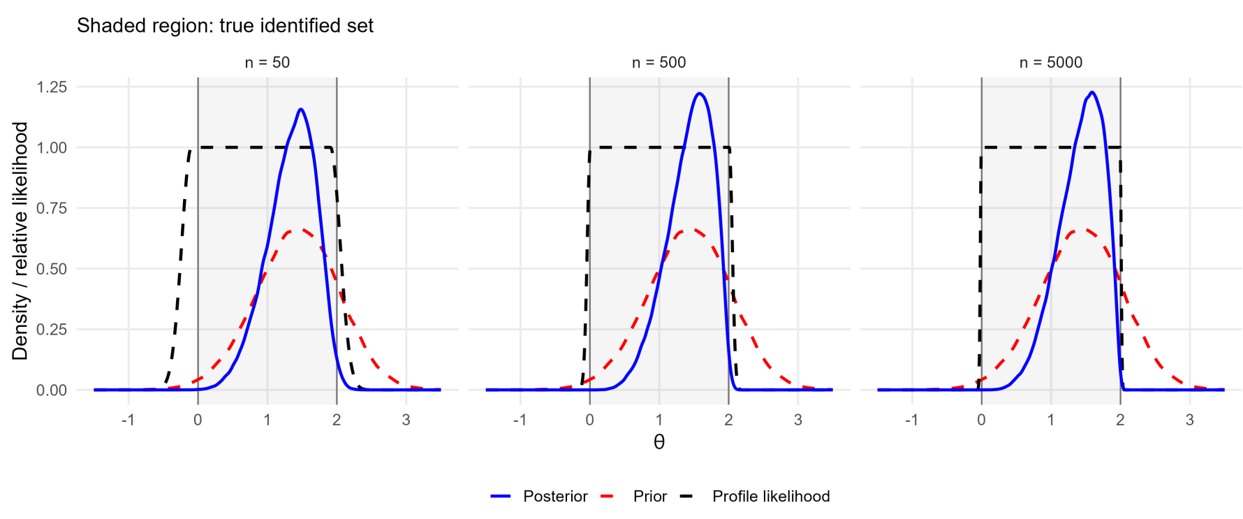
We can learn ϕ from the data:

More data, likelihood dominates



But conditional on ϕ, the likelihood cannot distinguish values of θ inside the identified set, $\pi(\theta \mid Y, \phi) = \pi(\theta \mid \phi)$.

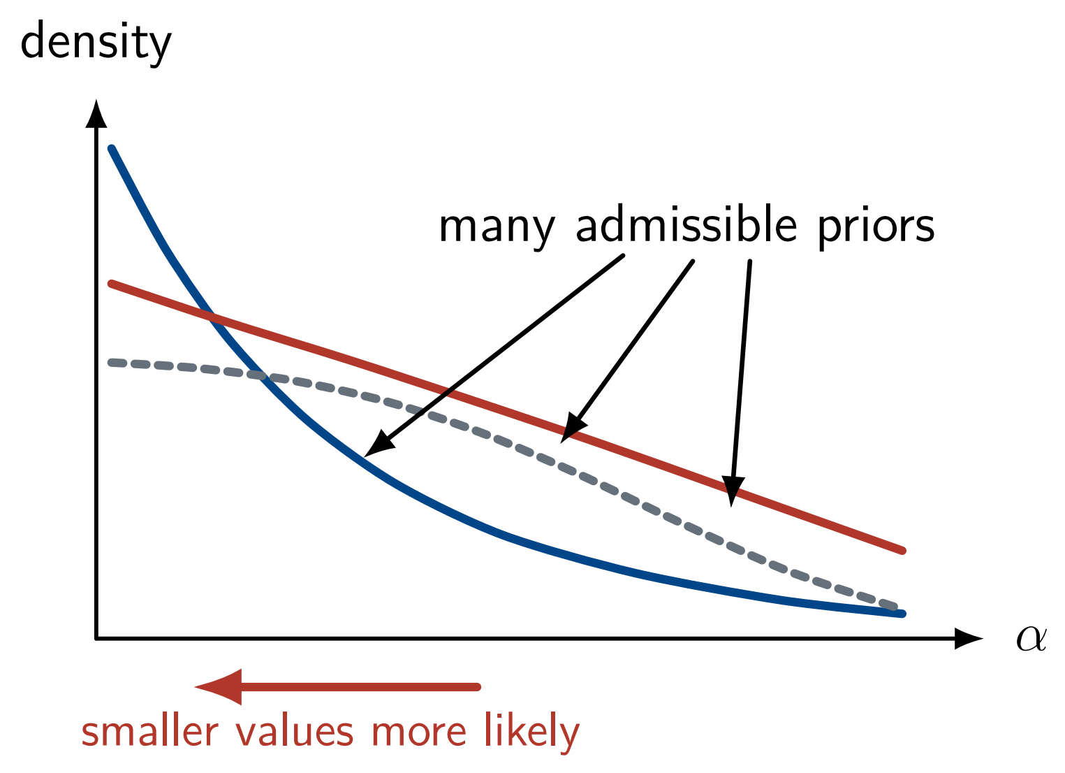
Conditional prior is never updated



> **From a Bayesian perspective, this is no problem if we have a precise prior.**

Economics does not identify prior

E.g. "smaller α is more likely" does not determine a unique prior.



$$\mathcal{M}_S = \left\{\pi_{\theta|\phi} : \text{weakly decreasing density over } \alpha\right\}.$$

Primitive: a class of marginals

For fixed ϕ, define

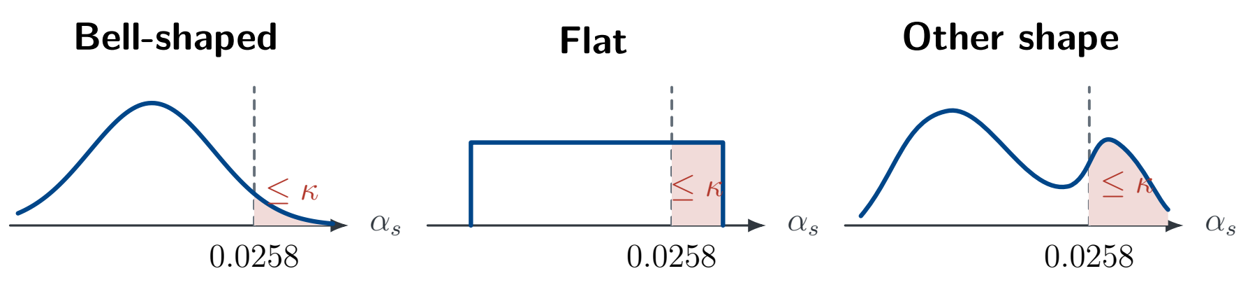
$$\Pi(\phi) = \left\{\pi_{\theta|\phi} \in \mathcal{P}(IS_\theta(\phi)) : S_{\#}\pi_{\theta|\phi} \in \mathcal{M}_S(\phi)\right\}.$$

The marginal class $\mathcal{M}_S(\phi)$ may encode event probabilities, moments, monotonicity or symmetry of its distribution.

For example, economic knowledge may only imply

$$P_\pi(S > c \mid \phi) \le \kappa.$$

This defines a class of priors

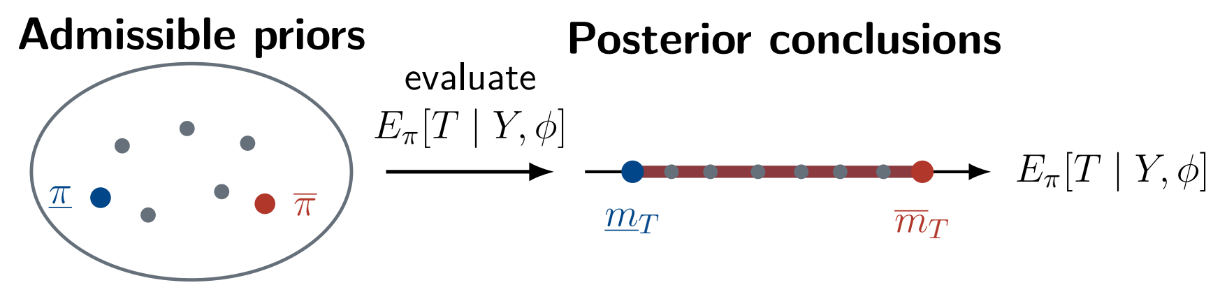


Same economic restriction, many admissible priors: only the probability of violating the tail event is controlled.

Defensible knowledge about $S(\theta)$ → **Admissible priors on θ** → **Posterior bounds for $T(\theta)$**

Population characterization

For a target $T : \Theta \to \mathbb{R}$, we want to bound posterior conclusions generated by the admissible priors:



$$\underline{m}_T(\phi) = \inf_{\pi \in \Pi(\phi)} E_\pi[T \mid Y, \phi]$$
$$\overline{m}_T(\phi) = \sup_{\pi \in \Pi(\phi)} E_\pi[T \mid Y, \phi]$$

In a generic setting

$$\mathbb{E}_{\tilde{\pi}}[T \mid Y, \phi] = \frac{\int T(\theta)\, p(Y \mid \theta, \phi)\, \tilde{\pi}_{\theta|\phi}(d\theta)}{\int p(Y \mid \theta, \phi)\, \tilde{\pi}_{\theta|\phi}(d\theta)}.$$

Nonlinear in prior. See Lavine, Wasserman, and Wolpert (1991)

But in set-identified models $p(Y \mid \theta, \phi) = p(Y \mid \phi)$ for all $\theta \in \mathrm{IS}_\theta(\phi)$. Hence

$$\mathbb{E}_{\tilde{\pi}}[T \mid Y, \phi] = \int T(\theta) \tilde{\pi}_{\theta|\phi}(d\theta).$$

i.e. $\tilde{\pi}_{\theta|\phi,Y} = \tilde{\pi}_{\theta|\phi}$.

Theorem 1 — Robust posterior bounds

Under standard regularity conditions,

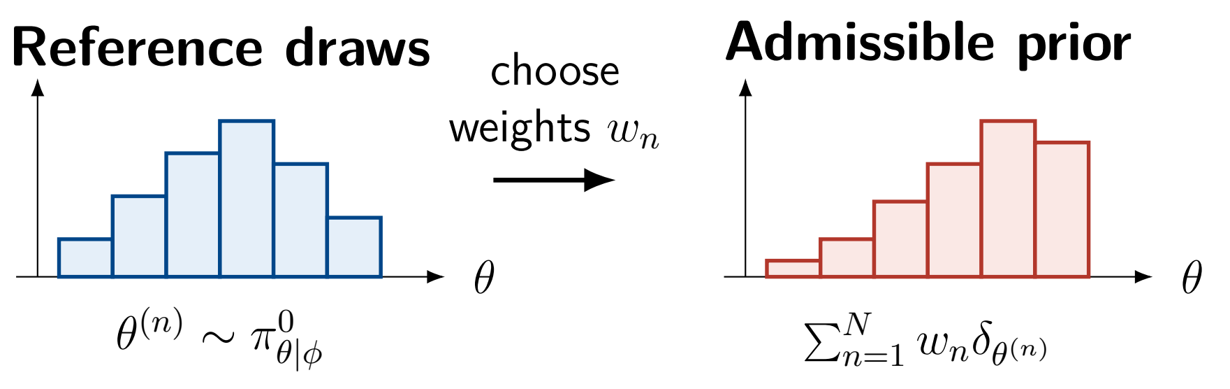
$$\underline{m}_T(\phi) = \min_{\pi \in \Pi(\phi)} \int T(\theta)\, \pi(d\theta)$$
$$\overline{m}_T(\phi) = \max_{\pi \in \Pi(\phi)} \int T(\theta)\, \pi(d\theta)$$

If the restrictions defining $\mathcal{M}_S(\phi)$ are linear, the bounds solve **infinite-dimensional linear programs**.

Approximation

Start with a reference prior $\pi^0_{\theta|\phi}$ and draw:

$$\theta^{(1)}, \dots, \theta^{(N)} \overset{iid}{\sim} \pi^0_{\theta|\phi}, \qquad \mathrm{supp}(\pi^0_{\theta|\phi}) = \mathrm{IS}_\theta(\phi).$$


Reference draws
θ
$\theta^{(n)} \sim \pi^0_{\theta|\phi}$
choose
weights w_n
Admissible prior
θ
$\sum_{n=1}^N w_n \delta_{\theta^{(n)}}$

Compute $S(\theta^{(n)}), T(\theta^{(n)})$ for all n and search over probability weights

$$w_n \ge 0, \qquad \sum_{n=1}^{N} w_n = 1.$$

The prior class $\mathcal{M}_S(\phi)$ constrains the feasible weights to $\mathcal{W}_N(\phi)$.

Theorem 2 — Finite approximation

The simulated robust bounds are the linear programs

$$\underline{m}_{N,T}(\phi) = \min_{w \in \mathcal{W}_N(\phi)} \sum_{n=1}^{N} w_n T(\theta^{(n)}),$$

Under standard regularity and support conditions,

$$\underline{m}_{N,T}(\phi) \to \underline{m}_T(\phi), \qquad \overline{m}_{N,T}(\phi) \to \overline{m}_T(\phi) \qquad \text{a.s.}$$

> **Prior sensitivity requires only solving finite dimensional linear programs.**

Oil-market application

Do conclusions in the oil market literature depend on prior specifications?

Kilian and Murphy (2014)-style hard restrictions include

$$0 \le \alpha_s \le 0.0258$$

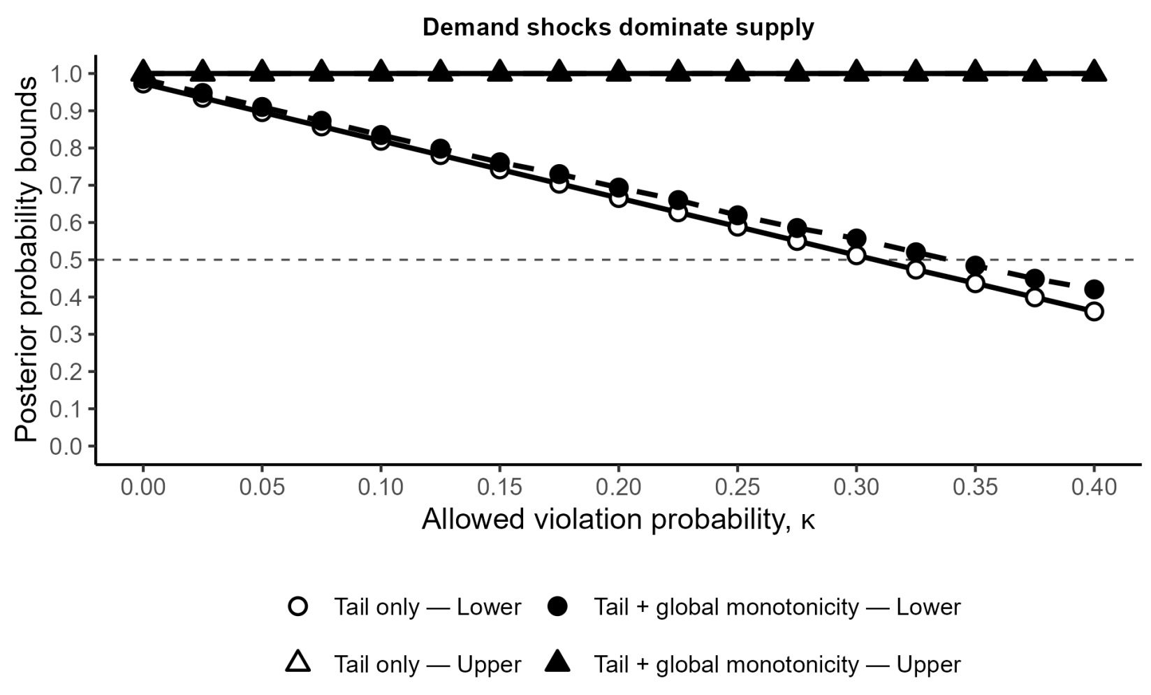
Replace them by classes of conditional priors defined only by violation probabilities:

$$P_\pi(\alpha_s^F > 0.0258 \mid \phi) \le \kappa,$$
$$P_\pi(\alpha_s^S > 0.0258 \mid \phi) \le \kappa$$

Prior information $S(\theta)$: elasticities → **Oil-market SVAR θ** → **Target $T(\theta)$: 12-month FEVD**

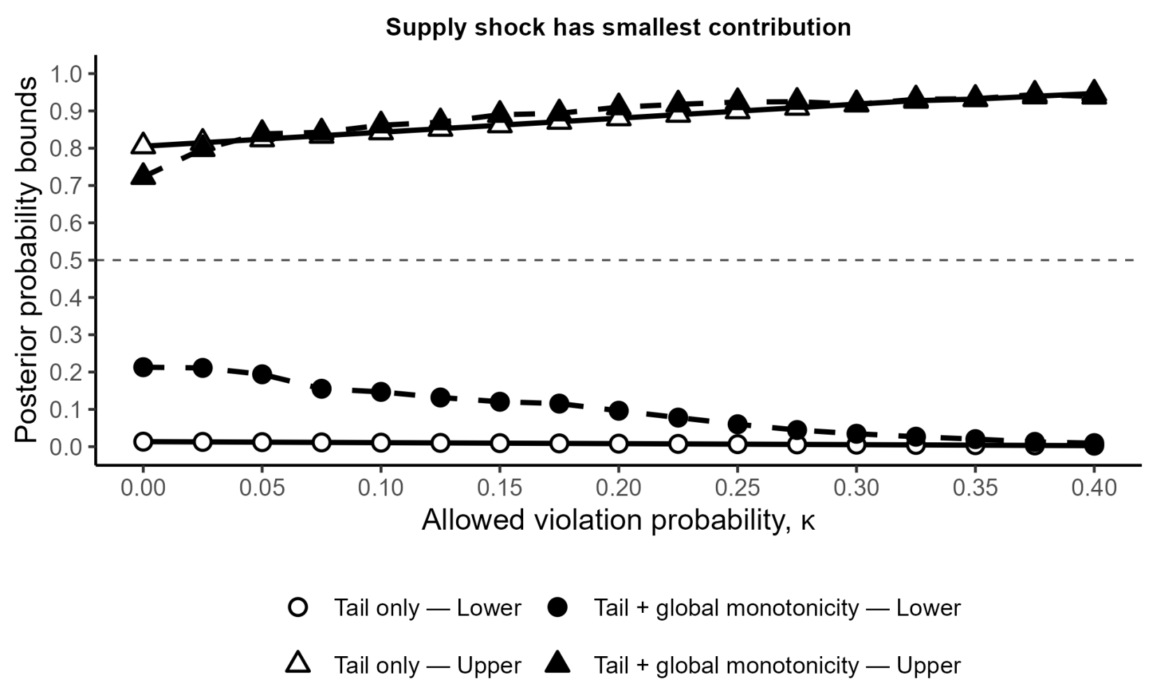
Demand dominance

$$\underline{p}(\kappa) = \inf_{\pi \in \Pi_\kappa} P_\pi(\text{FEVD}_{12}^{\text{demand}} - \text{FEVD}_{12}^{\text{supply}} > 0)$$
$$\bar{p}(\kappa) = \sup_{\pi \in \Pi_\kappa} P_\pi(\text{FEVD}_{12}^{\text{demand}} - \text{FEVD}_{12}^{\text{supply}} > 0)$$



Supply irrelevance

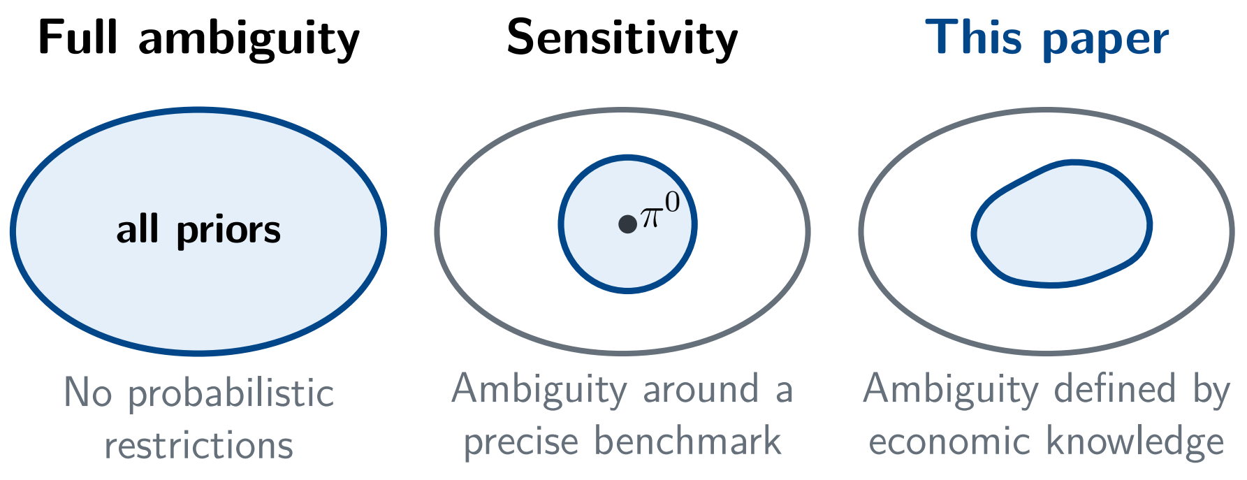
$$\underline{p}(\kappa) = \inf_{\pi \in \Pi_\kappa} P_\pi(\text{FEVD}_{12}^{\text{supply}} \text{is smallest shock})$$
$$\bar{p}(\kappa) = \sup_{\pi \in \Pi_\kappa} P_\pi(\text{FEVD}_{12}^{\text{supply}} \text{is smallest shock})$$



> **Demand dominance is robust, supply being least important is not.**

Related literature

My proposal builds prior classes based on marginal information we have rather than considering all priors or sensitivity around a full joint prior.


Full ambiguity
all priors
No probabilistic restrictions
Sensitivity
π^0
Ambiguity around a precise benchmark
This paper
Ambiguity defined by economic knowledge

Full ambiguity and extreme beliefs

In the Moon–Schorfheide example,

$$IS_\theta(\phi) = [\phi, \phi + 2].$$

Giacomini and Kitagawa (2021) allow *every* conditional prior supported on this interval. Hence the set of posterior means is

$$[\phi, \phi + 2].$$

But attaining the upper endpoint requires

$$E_\pi[\theta \mid Y, \phi] = \phi + 2 \qquad \Longrightarrow \qquad \pi_{\theta|\phi} = \delta_{\phi+2}.$$

In applications, we might feel confident excluding some priors.

For instance, if we only require lower values of θ to be weakly more likely,

$$\theta = \phi + 2U, \qquad f_U \text{ weakly decreasing on } [0, 1],$$

then set of posterior means spans $[\phi, \phi + 1]$

Selected references. Moon and Schorfheide (2012); Giacomini, Kitagawa, and Uhlig (2019); Giacomini and Kitagawa (2021); Ho (2023); Kilian and Murphy (2014); Baumeister and Hamilton (2015), Baumeister and Hamilton (2019), Lavine, Wasserman, and Wolpert (1991), Lam, Liu, and Singham (2024), Martin (2026), Balch, Martin, and Ferson (2019), Kessler, Hoff, and Dunson (2015) .

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